ZAINA RAHEEN A P

241801328

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns


df = pd.read_csv("/content/Sample - Superstore.csv",  encoding='latin1')

print("Dataset loaded successfully!")
print("Dataset Shape:", df.shape)


print("\nFirst 5 rows:")
print(df.head())

print("\nColumn Names:")
print(df.columns)

print("\nDataset Information:")
print(df.info())

print("\nMissing Values:")
print(df.isnull().sum())

print("\nStatistical Summary:")
print(df.describe())


df["Order Date"] = pd.to_datetime(df["Order Date"])
category_sales = (
 df.groupby("Category")["Sales"]
 .sum()
 .sort_values(ascending=False)
 )

print("\nSales by Category:")
print(category_sales)

plt.figure(figsize=(8, 5))

sns.barplot(
 x=category_sales.index,
 y=category_sales.values
 )

plt.title("Sales by Category")
plt.xlabel("Category")
plt.ylabel("Total Sales")

plt.tight_layout()
plt.show()

category_region = (
 df.groupby(["Category", "Region"])["Sales"]  .sum()
 .reset_index()
```

```
)

plt.figure(figsize=(10, 6))

sns.barplot(
 data=category_region,
 x="Category",
 y="Sales",
 hue="Region"
)

plt.title("Sales by Category and Region")
plt.xlabel("Category")
plt.ylabel("Sales")

plt.tight_layout()
plt.show()
df["Month"] = df["Order Date"].dt.to_period("M").astype(str)

monthly_sales = (
 df.groupby(["Month", "Region"])["Sales"]
 .sum()
 .reset_index()
)

plt.figure(figsize=(12, 6))

sns.lineplot(
 data=monthly_sales,
 x="Month",
 y="Sales",
 hue="Region",
 marker="o"
)

plt.title("Sales Trend Over Time")
plt.xlabel("Month")
plt.ylabel("Sales")

plt.xticks(rotation=45)

plt.tight_layout()
plt.show()

scatter_data = (
 df.groupby(
 ["Sub-Category", "Category"],
 as_index=False
 )[["Sales", "Profit"]]
 .sum()
)

print("\nProfit vs Sales:")
print(scatter_data)
plt.figure(figsize=(10, 7))
```

```
sns.scatterplot(
data=scatter_data,
x="Sales",
y="Profit",
hue="Category",
s=150
)
for _, row in scatter_data.iterrows():

plt.text(
row["Sales"],
row["Profit"],
row["Sub-Category"],
fontsize=9
)

plt.title("Profit vs Sales by Sub-Category")
plt.xlabel("Sales")
plt.ylabel("Profit")

plt.axhline(0, linestyle="--")

plt.tight_layout()
plt.show()
summary = df.groupby("Category")[["Sales", "Profit"]].sum()

print("\nSales and Profit by Category:")
print(summary)
def filter_by_region(region):

filtered_df = df[df["Region"] == region]

print("\nSelected Region:", region)

print("\nSales by Category:")
print(
filtered_df.groupby("Category")["Sales"] .sum()
.sort_values(ascending=False)
)

print("\nTotal Sales:",
round(filtered_df["Sales"].sum(), 2))

print("Total Profit:",
round(filtered_df["Profit"].sum(), 2))



filter_by_region("East")
def filter_by_category(category):

filtered_df = df[df["Category"] == category] print("\nSelected
```

```
Category:", category)

 print("Total Sales:",
 round(filtered_df["Sales"].sum(), 2))

 print("Total Profit:",
 round(filtered_df["Profit"].sum(), 2))  print("\nSales by Region:")

 print(
 filtered_df.groupby("Region")["Sales"]
 .sum()
 .sort_values(ascending=False)
 )
filter_by_category("Technology")
 total_sales = df["Sales"].sum()

total_profit = df["Profit"].sum()

print("\n" + "=" * 60)
print("SALES AND PROFIT DASHBOARD SUMMARY")
print("=" * 60)

print("Total Sales :", round(total_sales, 2))
print("Total Profit:", round(total_profit, 2))

print("\nTotal Sales by Region:")
print(
 df.groupby("Region")["Sales"]
 .sum()
 .sort_values(ascending=False)
 )

print("\nTotal Sales by Category:")
print(
 df.groupby("Category")["Sales"]
 .sum()
 .sort_values(ascending=False)
 )

print("\nTotal Profit by Category:")
print(
 df.groupby("Category")["Profit"]
 .sum()
 .sort_values(ascending=False)
 )

Dataset loaded successfully!
Dataset Shape: (9994, 21)

 First 5 rows:
 Row ID Order ID Order Date Ship Date Ship Mode  Customer ID \
0 1 CA-2016-152156 11/8/2016 11/11/2016 Second Class  CG-12520
 1 2 CA-2016-152156 11/8/2016 11/11/2016 Second Class  CG-12520
2 3 CA-2016-138688 6/12/2016 6/16/2016 Second Class  DV-13045
```

3 4 US-2015-108966 10/11/2015 10/18/2015 Standard Class SO-20335
4 5 US-2015-108966 10/11/2015 10/18/2015 Standard Class SO-20335

Customer Name Segment Country City ... \ 0 Claire Gute Consumer United States Henderson ... 1 Claire Gute Consumer United States Henderson ...
2 Darrin Van Huff Corporate United States Los Angeles ... 3 Sean O'Donnell Consumer United States Fort Lauderdale ... 4 Sean O'Donnell Consumer United States Fort Lauderdale ...

Postal Code Region Product ID Category Sub Category \
0 42420 South FUR-BO-10001798 Furniture Bookcases 1 42420 South FUR-CH-10000454

Furniture Chairs 2 90036 West OFF-LA-10000240 Office Supplies Labels 3 33311 South

FUR-TA-10000577 Furniture Tables 4 33311 South OFF-ST-10000760 Office Supplies

Storage

Product Name Sales Quantity \
0 Bush Somerset Collection Bookcase 261.9600 2
1 Hon Deluxe Fabric Upholstered Stacking Chairs,... 731.9400 3
2 Self-Adhesive Address Labels for Typewriters b... 14.6200 2
3 Bretford CR4500 Series Slim Rectangular Table 957.5775 5
4 Eldon Fold 'N Roll Cart System 22.3680 2

Discount Profit
0 0.00 41.9136
1 0.00 219.5820
2 0.00 6.8714
3 0.45 -383.0310
4 0.20 2.5164

[5 rows x 21 columns]

Column Names:
Index(['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode', 'Customer ID', 'Customer Name', 'Segment', 'Country', 'City', 'State',
'Postal Code', 'Region', 'Product ID', 'Category', 'Sub Category',
'Product Name', 'Sales', 'Quantity', 'Discount', 'Profit'], dtype='object')

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993 Data
columns (total 21 columns):
Column Non-Null Count Dtype --- ------ -------------- -----
0 Row ID 9994 non-null int64 1 Order ID 9994 non-null object
2 Order Date 9994 non-null object 3 Ship Date 9994 non-null object 4 Ship Mode 9994 non-null object 5 Customer ID 9994 non-null object 6 Customer Name 9994 non-null object 7 Segment 9994 non-null object 8 Country 9994 non-null object 9 City 9994 non-null object 10 State 9994 non-null object 11 Postal Code 9994

non-null int64
12 Region 9994 non-null object 13 Product ID 9994 non-null object 14 Category 9994 non-null object 15 Sub-Category 9994 non-null object 16 Product Name 9994 non-null object 17 Sales 9994 non-null float64 18 Quantity 9994 non-null int64
19 Discount 9994 non-null float64 20 Profit 9994 non-null float64 dtypes: float64(3), int64(3), object(15) memory usage: 1.6+ MB
None

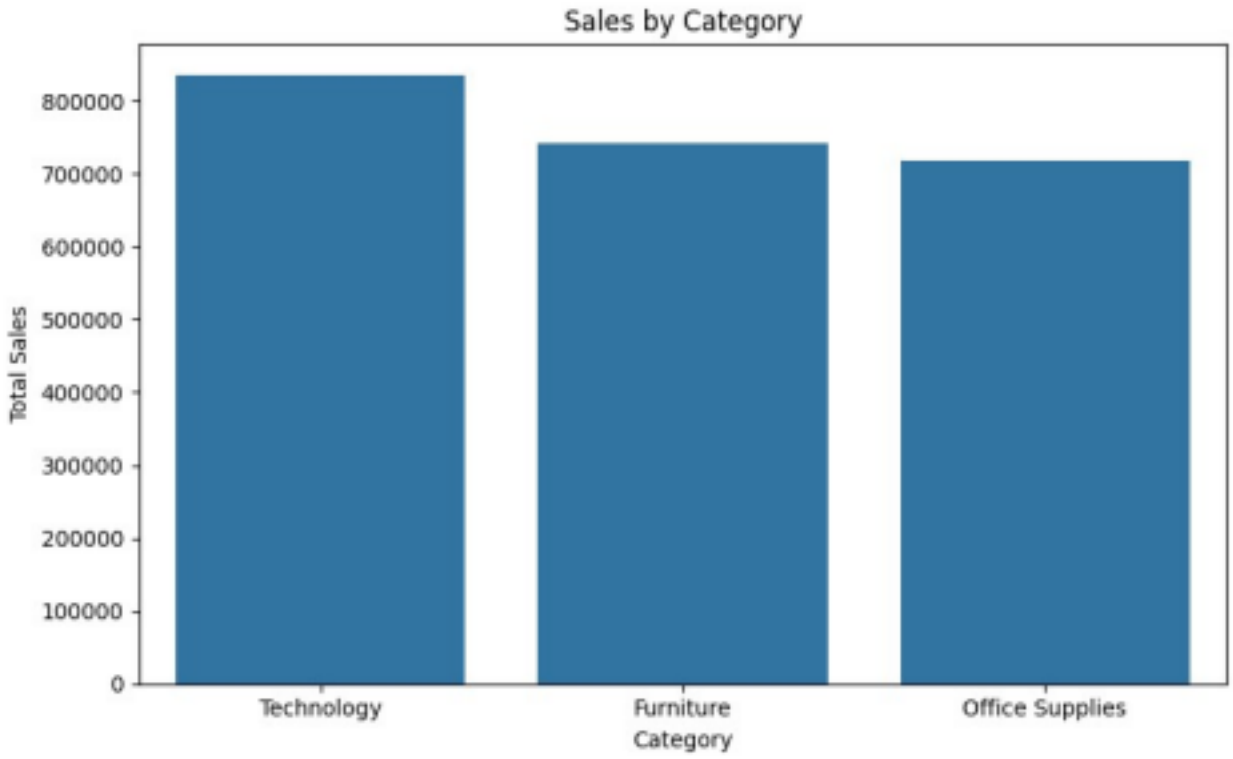
Missing Values:
Row ID 0
Order ID 0
Order Date 0
Ship Date 0
Ship Mode 0
Customer ID 0
Customer Name 0
Segment 0
Country 0
City 0
State 0
Postal Code 0
Region 0
Product ID 0
Category 0
Sub-Category 0
Product Name 0
Sales 0
Quantity 0

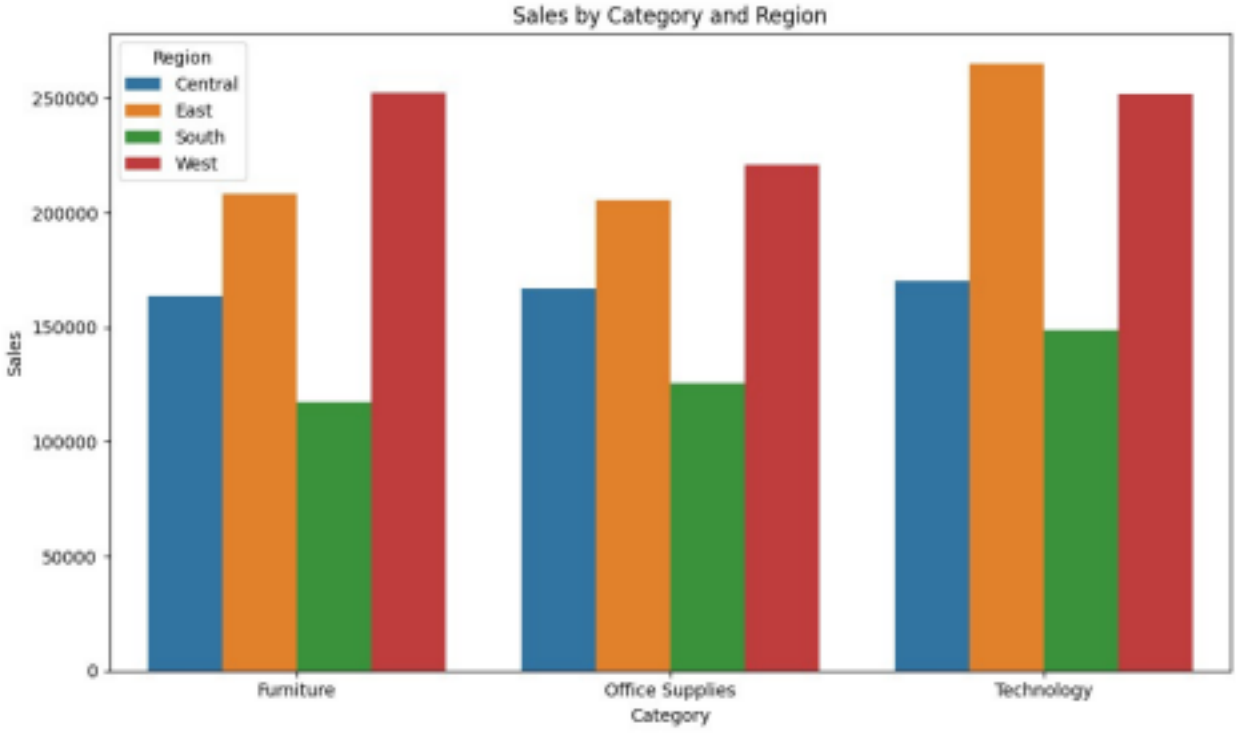
```
Discount 0
Profit 0
dtype: int64

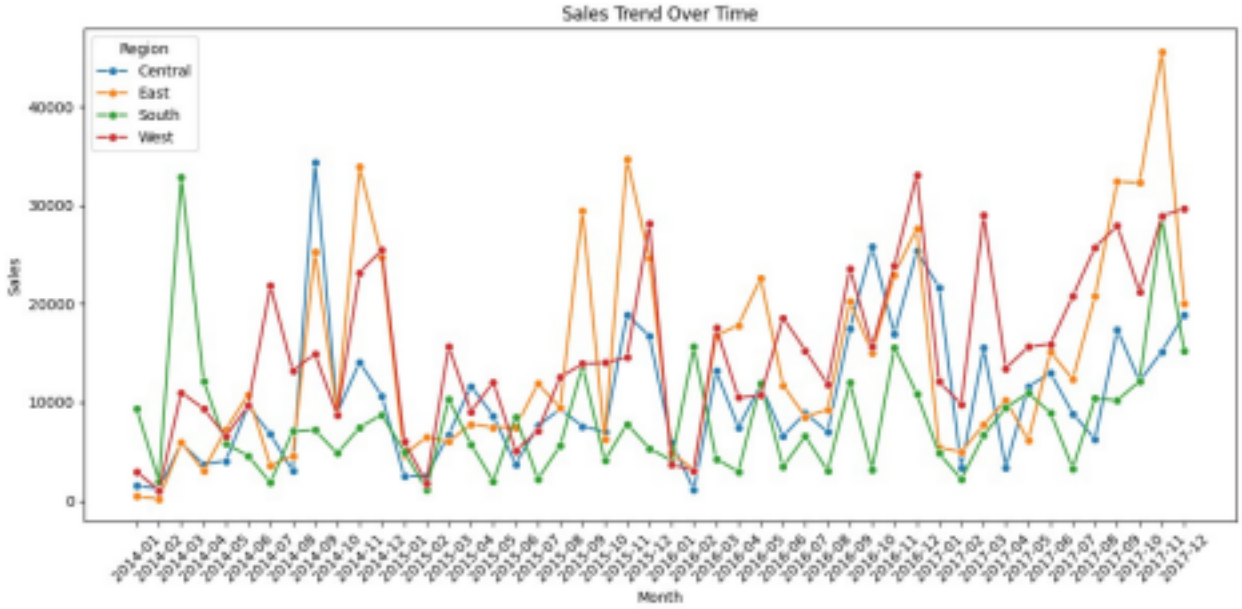
Statistical Summary:
 Row ID Postal Code Sales Quantity  Discount \
count 9994.000000 9994.000000 9994.000000 9994.000000  9994.000000
mean 4997.500000 55190.379428 229.858001 3.789574  0.156203
std 2885.163629 32063.693350 623.245101 2.225110  0.206452
min 1.000000 1040.000000 0.444000 1.000000  0.000000
25% 2499.250000 23223.000000 17.280000 2.000000  0.000000
50% 4997.500000 56430.500000 54.490000 3.000000  0.200000
75% 7495.750000 90008.000000 209.940000 5.000000  0.200000
max 9994.000000 99301.000000 22638.480000 14.000000  0.800000

 Profit
count 9994.000000
mean 28.656896
std 234.260108
min -6599.978000
25% 1.728750
50% 8.666500
75% 29.364000
max 8399.976000

Sales by Category:
Category
Technology 836154.0330
Furniture 741999.7953
Office Supplies 719047.0320
Name: Sales, dtype: float64
```

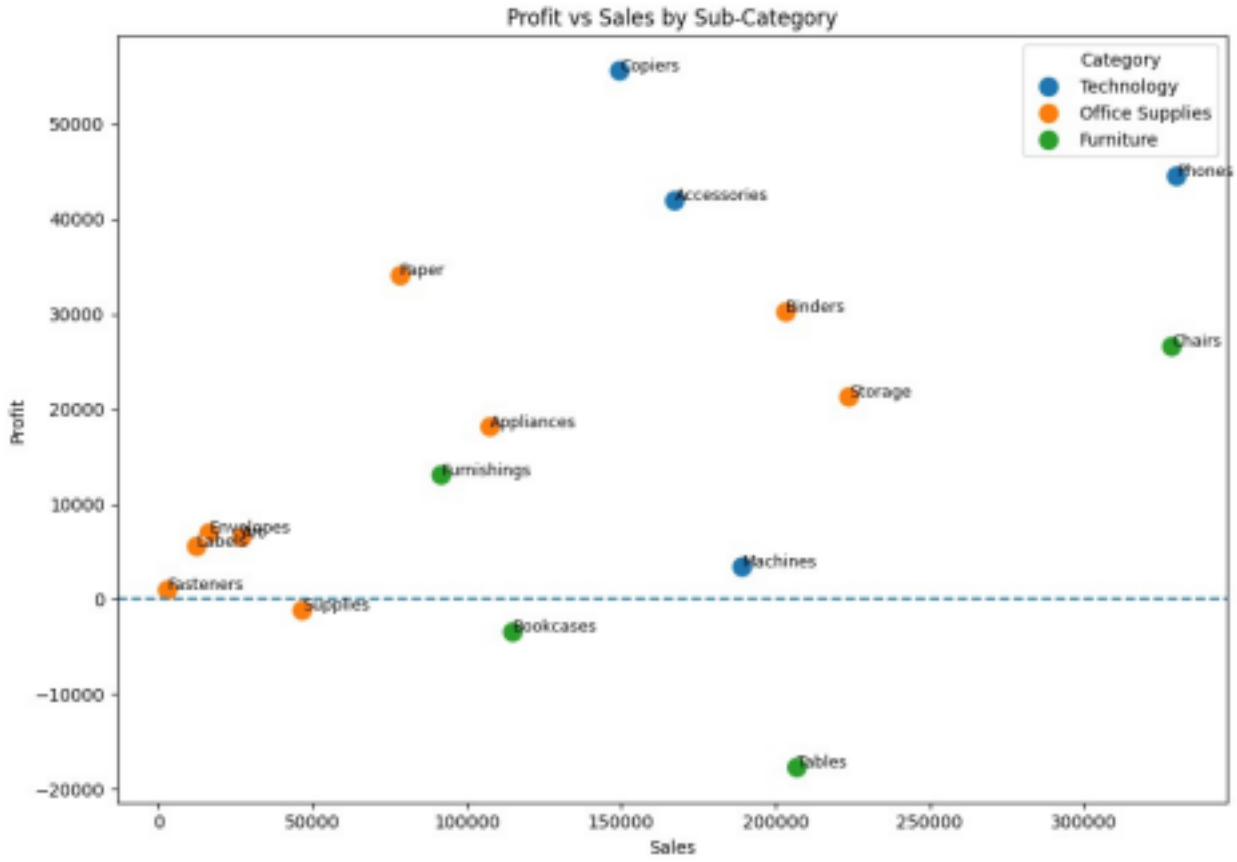

Sales by Category
Total Sales
800000
700000
600000
500000
400000
300000
200000
100000
0
Technology
Furniture
Office Supplies
Category


Sales by Category and Region
Region
Central
East
South
West
Sales
250000
200000
150000
100000
50000
0
Furniture
Office Supplies
Technology
Category



Profit vs Sales:

Sub-Category Category Sales Profit 0 Accessories Technology 167380.3180 41936.6357 1 Appliances Office Supplies 107532.1610 18138.0054 2 Art Office Supplies 27118.7920 6527.7870 3 Binders Office Supplies 203412.7330 30221.7633 4 Bookcases Furniture 114879.9963 -3472.5560 5 Chairs Furniture 328449.1030 26590.1663 6 Copiers Technology 149528.0300 55617.8249 7 Envelopes Office Supplies 16476.4020 6964.1767 8 Fasteners Office Supplies 3024.2800 949.5182 9 Furnishings Furniture 91705.1640 13059.1436 10 Labels Office Supplies 12486.3120 5546.2540 11 Machines Technology 189238.6310 3384.7569 12 Paper Office Supplies 78479.2060 34053.5693 13 Phones Technology 330007.0540 44515.7306 14 Storage Office Supplies 223843.6080 21278.8264 15 Supplies Office Supplies 46673.5380 -1189.0995 16 Tables Furniture 206965.5320 -17725.4811



Sales and Profit by Category:
Sales Profit
Category
Furniture 741999.7953 18451.2728
Office Supplies 719047.0320 122490.8008
Technology 836154.0330 145454.9481

Selected Region: East

Sales by Category:
Category
Technology 264973.981
Furniture 208291.204
Office Supplies 205516.055
Name: Sales, dtype: float64

Total Sales: 678781.24
Total Profit: 91522.78

Selected Category: Technology
Total Sales: 836154.03
Total Profit: 145454.95

```
Sales by Region:
Region
 East 264973.981
West 251991.832
Central 170416.312
South 148771.908
Name: Sales, dtype: float64

============================================================
SALES AND PROFIT DASHBOARD SUMMARY
============================================================
Total Sales : 2297200.86
Total Profit: 286397.02

Total Sales by Region:
Region
West 725457.8245
 East 678781.2400
Central 501239.8908
South 391721.9050
Name: Sales, dtype: float64

Total Sales by Category:
Category
Technology 836154.0330
 Furniture 741999.7953
Office Supplies 719047.0320
Name: Sales, dtype: float64

Total Profit by Category:
Category
Technology 145454.9481
Office Supplies 122490.8008
 Furniture 18451.2728
Name: Profit, dtype: float64
```